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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

EE3101-1 – POWER ELECTRONICS
(For EEE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. For a Power MOSFET, the main advantage over BJTs in high-frequency applications is:
A) Higher current gain
B) Absence of minority carrier storage effects
C) Lower conduction resistance at all current levels
D) Higher avalanche breakdown voltage
2. In an IGBT, the presence of the p+ substrate layer leads to:
A) Reduced switching losses compared to MOSFET
B) Higher forward conduction drop due to conductivity modulation
C) Elimination of latch-up during operation
D) Better avalanche ruggedness than MOSFET
3. A snubber circuit across a thyristor is designed to:
A) Reduce conduction losses in ON-state
B) Suppress harmonics in the load current
C) Limit dv/dt across the device during turn-OFF
D) Improve natural commutation capability of the device
4. A TRIAC is triggered into conduction by:
A) Only positive gate current applied at positive main terminal voltage
B) Only negative gate current applied at negative main terminal voltage
C) Suitable gate current pulses in all four quadrants of operation
D) Gate current applied only when MT1 is positive with respect to MT2
5. A single-phase full-wave phase-controlled converter with resistive load is fed from a 230 V (rms) sinusoidal source. If the firing angle $\alpha = 60^\circ$, What is V_{avg} (in volts)?
A) 115 V
B) 155.3 V
C) 200 V
D) 86.6 V
6. While paralleling two power MOSFETs for higher current, which practice best helps ensure stable current sharing during switching?
A) Use identical MOSFETs with no series elements
B) Put a small source-resistor (emitter/source balancing resistor) in each device and individual gate resistors
C) Increase gate drive amplitude to reduce $R_{ds(on)}$ spread
D) Rely on package thermal coupling only
7. A power diode exhibits reverse-recovery charge $Q_{rr} = 4.0 \mu C$ and the peak reverse recovery current $I_{rm} \approx 8.0 A$. A simple approximation for recovery time, t_{rr} is
A) 0.5 s
B) 0.5 ms
C) 0.5 μs
D) 0.5 ns

8. Which commutation method uses the AC supply voltage polarity reversal to turn-off a conducting SCR in an AC converter?
- A) Forced (auxiliary) commutation with LC circuit B) Resonant commutation using tuned network
- C) Natural (line) commutation — line voltage reversal provides reverse voltage D) Complementary (push-pull) commutation with another SCR
9. In a buck-boost converter, input voltage = 12 V, duty cycle $D=0.5$. The output voltage is:
- A) +12 V B) -12 V
- C) +24 V D) -24 V
10. In a buck converter, the average output voltage is directly proportional to:
- A) Load resistance B) Duty cycle
- C) Inductor value D) Switching frequency
11. In discontinuous conduction mode of a buck converter:
- A) Inductor current never becomes zero B) Switching frequency has no effect
- C) Output voltage is unaffected D) Inductor current becomes zero for part of the cycle
12. The polarity of output in a buck-boost converter relative to input is:
- A) Always negative B) Always positive
- C) Same as input D) Independent of duty cycle
13. In a single-phase full converter with RL load, a freewheeling diode is used to
- A) Reduce average output voltage B) Increase input power factor
- C) Provide a path for current when input reverses D) Reduce duty cycle
14. In a three-phase full converter, compared to a single-phase converter:
- A) Ripple frequency is lower and output unstable B) Output voltage is independent of firing angle
- C) Efficiency decreases D) Ripple frequency is higher and DC output smoother
15. The primary reason to operate converters in continuous conduction mode is
- A) To reduce inductor size B) To minimize output voltage ripple
- C) To reduce switching frequency D) To reverse the output polarity
16. In practical power supplies for computers and telecom equipment, a buck converter is preferred because:
- A) It reduces a higher DC bus voltage to regulated lower levels efficiently B) It can provide negative output voltage
- C) It works only in discontinuous mode D) It requires no filtering components
17. A single-phase inverter produces a square wave of peak 200 V. The rms value of the fundamental component is:
- A) 127 V B) 180 V
- C) 141 V D) 200 V
18. A three-phase bridge inverter is supplied with 400 V DC. The fundamental line voltage RMS is approximately
- A) 180 V B) 462 V
- C) 254 V D) 400 V
19. The main advantage of multilevel inverters over conventional two-level inverters is:
- A) Lower cost B) Higher switching losses
- C) Elimination of control circuits D) Reduced harmonic distortion and improved voltage quality
20. In resonant inverters (Class D or Class E), the key advantage is:
- A) High efficiency due to soft switching B) Independent of load variations
- C) Higher output frequency with no harmonics D) Elimination of passive components

PART – B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the V-I characteristics of an SCR with neat sketches.	5	L2	1	1,2
	b) With the help of a diagram, explain the switching characteristics of a Power MOSFET during turn-on and turn-off transitions.	5	L2	1	1,2
	c) List three types of power electronics circuits and explain briefly.	6	L1	1	1,2
2.	a) Draw the equivalent two-transistor model of an SCR and explain its operation.	5	L*2	1	1,2
	b) The parameters of UJT are $V_s=30V$, $\eta=0.51$, $I_p=10Ma$, $V_v=3.5V$ and $I_v=10Ma$. The frequency of oscillation is $f=50Hz$ and the width of the trigger pulse is $t_g=50\mu s$. Assume $V_D=0.5$. Calculate the corresponding components.	5	L*3	1	1,2
	c) Explain the static and dynamic characteristics of an IGBT with neat sketches.	6	L*2	1	1,2
3.	a) What are the specifications of power semiconductor switches? List and explain any five important specifications.	5	L*2	1	1,2
	b) Explain the peripheral effects in power electronic circuits, such as EMI and harmonics.	5	L*2	1	1,2
	c) The single-phase full-wave controller supplies an RL load. The input rms voltage is $V_s = 120 V$, 60 Hz. The load is such that $L = 6.5 Mh$ and $R = 2.5\Omega$. The delay angles of thyristors are equal: $\alpha_1 = \alpha_2 = \pi/2$. Determine (a) the conduction angle of thyristor T1, ; (b) the rms output voltage V_o , (c) the rms thyristor current I_R ; (d) the rms output current I_o ; (e) the average current of a thyristor I_A ; and (f) the input PF.	6	L*3	2	1,2
Unit – II					
4.	a) List any five design considerations for a buck converter and its filter.	5	L*2	3	1,2,3
	b) A buck converter has: $V_{in}=40, V_o=20, R=10\Omega$, Switching frequency $f_s=25kHz$ Inductor $L=200Mh$. (a) Calculate the duty ratio and average inductor current. (b) Determine the inductor current ripple and state whether the operation is CCM or DCM.		L*3	3	1,2,3
	c) With neat circuit diagram and waveforms explain the working of single phase full wave converter with RL load.	6	L*2	4	1,2,3
5.	a) With the help of circuit diagram and waveform, explain the principle of operation of a Boost Converter.	5	L*2	3	1,2,3
	b) Define a controlled rectifier. Explain the working of Three phase full-converter with suitable circuit sketches.	5	L*2	4	1,2,3
	c) A three-phase half wave converter is operated from a star connected 210V, 50Hz supply and the load resistance is 10Ω . If the average output voltage is 50% of maximum possible average output voltage, calculate i) Delay angle ii) The RMS and average output current iii) The RMS and average thyristor current iv) The input power factor	6	L*3	4	1,2,3

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|-------|---|---|-----|---|-------|
| 6. a) | Discuss briefly inductor current ripple and capacitor voltage ripple in Buck-Boost converter | 5 | L*2 | 3 | 1,2,3 |
| b) | If a single-phase controlled converter has a pure resistive load of $R\Omega$ and delay angle of $\pi/2$ (90°). Determine
i) Rectification efficiency
ii) Form factor
iii) Ripple factor
vi) Transformer utilization factor | 5 | L*3 | 4 | 1,2,3 |
| c) | Distinguish between LVR and SMPC | 6 | L*4 | 3 | 1,2,3 |

Unit – III

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|-------|---|---|-----|---|-------|
| 7. a) | State the differences between 180° conduction mode and 120° conduction mode in a three-phase bridge inverter. | 5 | L*4 | 5 | 1,2,3 |
| b) | Explain the operation of a variable DC-link inverter. | 5 | L*2 | 5 | 1,2,3 |
| c) | A single-phase full-bridge inverter has a resistive load of $R=5\Omega$ and the dc input voltage of $V_s=48$ volts. Determine its output power, average and peak current of each transistor, Total harmonic distortion. | 5 | L*3 | 5 | 1,2,3 |
| 8. a) | Draw the circuit diagram of a three-phase bridge inverter with Y connected resistive load. Sketch the gating signals and line to line output voltage for 120° conduction operation. | 5 | L*2 | 5 | 1,2,3 |
| b) | Define Harmonic factor, total harmonic distortion, distortion factor and lowest order harmonic. | 5 | L*1 | 5 | 1,2,3 |
| c) | Explain the SPWM technique in inverters with waveforms. | 6 | L*2 | 5 | 1,2,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
